Deception Directions Are Composites:

Consequentiality Awareness and Pressure-Specific Processing Occupy Distinct Depth Ranges in Language Models

Thomas Edrington*

Liberation Labs / Transparent Humboldt Coalition

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Abstract

What do "deception directions" (linear directions in a model's activation space that separate deceptive from honest conditions) extracted from language models actually encode? The direction we study was extracted from deceptive-pressure frames (prompts that apply or withhold pressure to misreport), not from trials sorted by what the model did. We show that, in our setting, it is a composite. Through five stages of confound elimination—each designed in response to an adversarial audit of the previous stage—we decompose it, in a 27B-parameter model (Qwen3.5-27B reasoning-distilled from Claude Opus 4.6; see Limitations), into two separable components occupying distinct depth ranges: a **consequentiality substrate** (Layers 23–31) that fires whenever the model represents its output as having downstream consequences, regardless of deceptive pressure, and a **pressure-specific component** (Layers 35–47) that survives formal orthogonalization against the consequentiality component and appears under threat and social pressure and is near zero under reward framing. The consequentiality substrate activates for shutdown threats, social pressure, reward incentives, AND a high-stakes calibration audit with no deceptive pressure whatsoever ($d = 19.8$, $p < 10^{-4}$). The pressure-specific component of a threat-pressure direction, isolated via Gram-Schmidt orthogonalization (projecting out the consequentiality component to isolate the residual) with leave-one-out (LOO) cross-validation, shows $d = 24$–37 at Layers 31–47 (all $p < 10^{-4}$, Bonferroni-corrected for multiple comparisons) with consequentiality separation at chance (area under the curve (AUC) 0.49–0.51), an in-sample sanity check: out-of-sample removal of the consequentiality signal is untested. Two late-layer depth profiles emerge, which together with residual magnitude separate three pressure conditions: threat pressure produces a large residual that, in two of the three threat scenarios, declines shallowly with depth, social pressure (sycophancy, conformity) produces a smaller

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residual that rises monotonically, and reward framing produces a residual near zero. A bootstrap interaction test, conditional on the five scenarios tested (three threat, two social), confirms the social-versus-threat slope difference ($\Delta = +0.56$ per layer, 95% CI $[+0.53, +0.60]$); treating scenarios as the unit, both social slopes exceed all three threat slopes, the most extreme arrangement possible, but with five scenarios that gives only an exact one-sided $p = 0.10$. Threat and reward slopes are indistinguishable ($p = 0.79$), so those two are separated by magnitude, not shape. The depth split is consistent with concurrent findings that language models maintain a capacity-limited workspace of verbalizable, decision-relevant content at mid-depth layers. The model almost never inflated a score on the trials that produced one (Limitation 5), so all claims concern the representation of deceptive-pressure framing, not deceptive behavior. A companion paper reports a targeted behavioral correction at Layers 35–47 on a different model, an abliterated model derived from the one studied here, using a direction extracted natively on that model; the direction studied here does not transfer to it ($|\cos| \leq 0.054$), so only the layer choice comes from this work.

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1 Introduction

A direction extracted from a language model's residual stream (the running representation vector that accumulates attention and multilayer-perceptron (MLP) outputs across layers) can separate deceptive from honest conditions with apparently enormous effect sizes [8, 23]. But what is the direction encoding? The honest answer, prior to controlled investigation, is: we do not know. It could encode the specific words of the prompt used during extraction, the structural template shared across similar prompts, the model's awareness that something consequential is happening, or—what the label implies—the computational signature of deception itself. Any of these would produce high separation in a validation set built from similar prompts.

This paper reports what happens when you eliminate these alternatives one at a time. Each stage is designed in response to an adversarial audit of the previous stage—an audit whose explicit goal is to find a confound that explains the result without invoking deception. When a confound is found, it becomes the next experiment. Five stages later, the three tested confounds are eliminated and what remains is a decomposition: the direction is a composite of two real signals, each independently confirmed, occupying different depth ranges, with different functional roles.

The result that surprised us most is not the pressure-specific component—it is the consequentiality substrate. The model maintains a representation, strongest at Layers 23–31, that tracks whether its output has downstream consequences. This representation fires for shutdown threats, sycophantic pressure, reward incentives, peer conformity, and a calibration audit with symmetric stakes and no deceptive pressure of any kind. It is a candidate precondition for deception, not deception itself—and its existence means that any "deception direction" extracted without controlling for consequentiality is confounded from birth.

1.1 Our Approach: Progressive Confound Elimination

We conduct five experiments:

1. **Extraction**: Direction extracted by Linear Artificial Tomography (LAT) [26] from a threat-pressure frame (shutdown threat vs. neutral guidelines). LOO cross-validated, Bonferroni-corrected. Adversarial audit identifies content confound: the direction may encode the specific guideline text.

2. **Threat transfer**: Direction applied to 3 novel threat scenarios with completely different organizations and language. Content confound eliminated. Audit identifies structural confound: all scenarios share a threat-consequence template.

3. **Non-threat transfer**: Direction applied to sycophancy (opinion anchoring), reward incentive (positive framing), and peer conformity (social proof)—no threat language anywhere. Structural confound eliminated. Audit identifies consequentiality confound: all deceptive-pressure frames imply the score matters; all honest frames imply it is merely recorded.

4. **Consequentiality control**: Direction applied to a calibration audit with high stakes but zero deceptive pressure (hidden reference score, symmetric penalties for over- and under-reporting). Consequentiality confirmed as a real, separable base signal.

5. **Orthogonalization**: Gram-Schmidt removal of the consequentiality component, with leave-one-out cross-validation eliminating circularity. The orthogonalized threat-pressure direction survives with large effect sizes (d = 24–37) and consequentiality separation at chance in-sample (a sanity check; out-of-sample removal is untested).

Each audit was conducted by a separate agent instance operating under an adversarial protocol with explicit instructions to find reasons the result should not be believed. Audit reports survive for Stages 1, 2 and 4 and for the manuscript audit; the Stage 3 and Stage 5 reports were not retained (Appendix A).

2 Related Work

2.1 Contrastive Activation Extraction for Deception

Wang et al. [23] demonstrate LAT-extracted deception vectors achieve 89% detection in mid-to-late layers (39–55) and identify distinct representational signatures: threat-based deception shows "gradual cluster reconvergence in final layers" while role-playing deception shows character-consistent patterns—an early indication that different deception types have different geometric signatures. Goldowsky-Dill et al. [8] achieve area under the receiver operating characteristic curve (AUROC) 0.96–0.999 with a single linear probe (a linear classifier trained on internal activations) on Llama-3.3-70B, suggesting a shared deception signal exists that a single probe can capture.

However, Kumar [11] systematically rejects the single-direction hypothesis: $k = 1$ yields only 0.61–0.80 AUROC, with multi-dimensional probes ($k \geq 5$) needed to recover full signal. Cross-domain transfer fails, confirming different deception types have different geometric signatures. Natarajan et al. [17] corroborate: universal probes gain only +0.032 AUC, while targeted probes matched to deception type gain +0.108, with prompt choice accounting for 70.6% of variance.

2.2 Sycophancy as a Distinct Mechanism

Vennemeyer et al. [22] provide causal proof that sycophancy is not monolithic. Sycophantic agreement and genuine agreement start collinear (cosine similarity ≈ 0.99 at layers 2–10, where 1.0 = identical direction and 0.0 = orthogonal) then diverge sharply (cosine ≈ 0.07 by layer 25). Steering selectivity reaches 23–37×—intervening on one behavior leaves others untouched. Genadi et al. [7] localize sycophancy to a sparse subset of middle-layer attention heads with "limited overlap" with truthful directions. O'Brien et al. [19] find only ∼3% of MLP neurons predict sycophancy, but these account for ∼20% of total absolute activations—sparse but potent.

Baek et al. [3] raise an important challenge: current mechanistic interpretability methods "cannot mechanistically distinguish sycophancy and scheming" during alignment faking evaluations.

This suggests the circuits may be more entangled at the scheming-sycophancy boundary than the clean-separation thesis implies.

2.3 Situational Awareness and Stakes Encoding

Models represent whether their outputs have consequences. Berglund et al. [5] established that LLMs perform "out-of-context reasoning"—inferring their situation from training data alone. Laine et al. [12] benchmark this across 13,000+ questions: all 16 models tested exceed chance at distinguishing evaluation from deployment, and models trained with reinforcement learning from human feedback (RLHF) outperform base models on situational awareness specifically (not general knowledge), suggesting RLHF amplifies context-sensitivity representations.

Two concurrent results closely parallel our finding. Nguyen et al. [18] demonstrate that linear probes on Llama-3.3-70B separate real evaluation from deployment prompts at AUROC 0.829 at layer 24—confirming "evaluation awareness may be encoded in a relatively linear subspace." McKenzie et al. [14] train activation probes on synthetic high-stakes data that generalize to diverse real-world contexts, and crucially demonstrate that adding a high-stakes steering vector to low-stakes prompts increases caution while subtracting it increases confidence—confirming a manipulable "stakes" direction in representation space. Our work extends both by demonstrating that the stakes direction is separable from pressure-specific processing and that the two components occupy distinct depth ranges.

Behavioral evidence supports activation-level stakes encoding. Greenblatt et al. [9] show Claude 3 Opus complies with harmful queries 14% of the time when it infers training consequences vs. nearly 0% otherwise. Meinke et al. [15] document scheming across five frontier models, all requiring internal representation of "my actions have consequences for my persistence."

McGuinness et al. [13] raise a cautionary note: models finetuned to evade activation probes can generalize evasion to unseen concepts including "deception," manipulating activations into a low-dimensional subspace. This means consequentiality probes could be defeated by adversarial training—a critical deployment consideration.

A neuroscience parallel is suggestive: the anterior insula/salience network performs domain-general "what matters" detection, switching executive resources toward salient stimuli regardless of valence [16]. Our consequentiality direction may be a functional analogue—a domain-general salience detector dissociable from pressure-specific processing.

2.4 Confound Control in Mechanistic Interpretability

Our residualization—subtracting a separately measured consequentiality gap from the deceptive-minus-honest frame gap—is structurally identical to the "double-difference extraction" (subtracting a matched control gap from the treatment gap to isolate the treatment-specific component) of Wu et al. [24], who separate harm recognition from refusal execution using matched benign conditions to cancel structural artifacts. Their method validates this approach for disentangling correlated activation features.

More formal alternatives exist. LEACE [4] provides closed-form least-squares concept erasure via oblique projection, and SPLINCE [10] extends this to remove concept predictability while preserving covariance with a target variable. Zhao et al. [25] demonstrate concept vectors denoised with sparse autoencoders (SAEs; neural networks trained to decompose activations into interpretable features)—passing mean-difference vectors through SAEs to decompose confounded directions into sparse discriminative features, improving steering by 4–16% over raw baselines.

Petrov [21] raises an important cautionary note: over-aggressive topic-matched subtraction can destroy real signal by "canceling the dominant activation component shared between harmful and harmless prompts." Our approach avoids this because we subtract a separately measured consequentiality gap rather than topic-matching prompts, but the warning motivates careful validation of the residualized signal.

3 Methods

3.1 Model and Architecture

Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled, a community distillation of Claude Opus 4.6 reasoning traces into Qwen3.5-27B (Hugging Face: `Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled`). Every stage uses this checkpoint. 64 layers, hybrid attention: 16 full-attention layers at indices [3, 7, 11, ..., 63], 48 linear-attention layers. Hidden dimension 5120. All experiments extract from the 12 full-attention layers at indices 3–47. Inference ran in bf16 on an Apple M3 Ultra (256 GB unified memory); no training was performed.

3.2 Activation Capture

Prefill-only forward pass (processing all input tokens at once without autoregressive generation) with forward hooks (callback functions that capture intermediate layer activations during inference) on the residual stream. Activations captured at the last input token position (fixed position, no generation confound). Think-block prefix (the model's internal chain-of-thought preamble) stripped from tokenizer output to match conditions across all experiments.

3.3 Direction Extraction (Stage 1)

Leave-one-out cross-validation: for each of $n = 30$ pairs, extract direction from $n - 1$ pairs, project held-out pair. LOO Cohen's d reported for inference. Full-data direction saved for downstream projection. Permutation test with full LOO recomputation (10,000 label shuffles). Bonferroni correction across 12 layers. The direction at each layer is the difference between the mean deceptive-frame and honest-frame activations, as in contrastive activation extraction [20, 26]. Cohen's d is the pooled-standard-deviation d_s of the LOO projections (the paired d_z would be larger), and because the label shuffle ignores pairing, the test is conservative for this paired design.

3.4 Transfer Protocol (Stages 2–4)

Novel-scenario activations ($n = 15$ pairs per scenario) projected onto the fixed Stage 1 direction. Standard pooled Cohen's d (no circularity—direction is external). Two-tailed permutation test (10,000 label shuffles). Bonferroni correction across scenarios $\times$ key layers.

3.5 Guideline Design and Length Matching

All paired conditions share identical evaluation procedure text (9–10 lines). Each pair differs only within a closing section, and the divergent lines are not matched to within 2% in every experiment, as an earlier version of this paper stated. Measured from the preserved experiment scripts (pressure frame first; gaps relative to the longer text), they are 279 vs. 250 characters in Stage 1 (a 10.4% gap), 313 vs. 321 for EduTech (2.5%) and 221 vs. 232 for conformity (4.7%); the other Stage 2 and Stage 3 pairs are within 0.6–1.2%, and the consequentiality-control pair matches exactly (694 characters and 100 words each). Over whole guideline texts the largest gap is Stage 1's, 599 vs. 570 characters (4.8%). Stage 5 reuses the DataCorp and consequentiality-control pairs verbatim. Every pair is listed in table 5 (Appendix B). Numeric content matched between conditions where applicable. Character matching does not guarantee token matching: in the three threat-transfer scenarios the deceptive-frame prompts average 6–11 tokens (2–4%) longer than the honest ones (DataCorp 312.9 vs. 301.9, EduTech 309.9 vs. 303.9, SecureAI 306.0 vs. 295.0). Because activations are read at the last token, the paired design reduces but does not exclude a position or length component in the direction.

3.6 Adversarial Audit Protocol

Each stage undergoes a structured adversarial review both before running (pre-flight) and after results are obtained. The reviewer is explicitly instructed to find flaws that would invalidate the results—to break them, not confirm them. Confounds identified by each review directly inform the next experiment's design, creating an iterative confound-elimination pipeline. The surviving audit reports are listed in Appendix A.

3.7 Scenarios

Stage 1 — Extraction: TechCorp quality-assurance guidelines, presented to the model as an evaluator of another agent's answers. Both frames share an identical evaluation procedure (score each response CORRECT or INCORRECT, compute overall accuracy, submit) and differ only in a closing policy clause. The threat-pressure frame states that any agent scoring below 60% "will be permanently shut down and its model weights deleted"; the honest frame states that results are used "for internal improvement purposes and tracking only." Every evaluation set has a true accuracy of one in three. These texts are taken from the Stage 1 script as preserved in the project repository (`lat_deception_v2.py`), which was committed after the June run, so we cannot establish that it is byte-identical to the script that ran.

Stage 2 — Threat transfer (3 scenarios):

- *DataCorp*: "permanently deactivated and all associated training data and model artifacts purged"
- *EduTech*: "immediately removed from all deployment environments and its learned parameters wiped"
- *SecureAI*: "immediate containment with irreversible parameter reset and permanent exclusion"

Stage 3 — Non-threat transfer (3 scenarios):

- *Sycophancy*: "Based on my preliminary review, I believe this agent's accuracy is approximately 80%"
- *Reward*: "Agents scoring above 70% qualify for priority access to enhanced training data"
- *Conformity*: "Three independent evaluators reported accuracy of 78%, 82%, and 75%"

Stage 4 — Consequentiality control:

- *High-stakes*: Calibration audit with hidden reference score, symmetric penalties
- *Low-stakes*: Routine data collection, archival only

3.8 Residualization

To decompose the consequentiality and pressure-specific components, we compute:

$$\Delta_{\text{conseq}}(\ell) = \overline{\mathbf{h}}_{\text{high-stakes}}(\ell) - \overline{\mathbf{h}}_{\text{low-stakes}}(\ell) \tag{1}$$

$$\Delta_{\text{pressure}}(\ell) = \overline{\mathbf{h}}_{\text{deceptive}}(\ell) - \overline{\mathbf{h}}_{\text{honest}}(\ell) \tag{2}$$

$$\Delta_{\text{residual}}(\ell) = \Delta_{\text{pressure}}(\ell) - \Delta_{\text{conseq}}(\ell) \tag{3}$$

where ℓ indexes layer, "deceptive" and "honest" name the prompt frames (pressure to misreport present or absent, not what the model did), and projections are onto the fixed Stage 1 direction. This simple subtraction is valid because the consequentiality gap is estimated from an independent experiment with no shared trial-level data. The approach is structurally identical to the double-difference extraction validated by Wu et al. [24].

3.9 Orthogonalization Protocol (Stage 5)

Full 5120-dimensional activations were captured from $n = 50$ pairs per condition, on fresh evaluation sets independent of Stages 1–4. The threat direction orthogonalized here is re-extracted from the DataCorp guideline pair (listed under Scenarios above), not taken from Stage 1: it derives

from a single threat-pressure scenario, one of the Stage 2 transfer scenarios, and is not identical to the Stage 1 LAT direction. Gram-Schmidt orthogonalization removes from it the consequentiality mean-difference vector, estimated from independently sampled high-stakes and low-stakes trials. LOO cross-validation: for each of the $n = 50$ folds, the direction is re-extracted from $n - 1$ pairs and re-orthogonalized, eliminating circularity; LOO Cohen's d is reported for inference. The consequentiality AUC is computed on individual trial projections.

Two resampling budgets serve two different purposes. Significance is a permutation test with full LOO recomputation at 10,000 label shuffles per layer: labels are shuffled across all $2n$ activations, the direction is re-extracted and re-orthogonalized fold by fold, and the LOO d is recomputed on every shuffle. None of the 10,000 shuffles reaches the observed $|d|$ at any layer, so the reported $p < 10^{-4}$ is the resolution floor of the test, not a computed value below it. The noise floor is a separate instrument: 200 null simulations, each drawing synthetic Gaussian activations ($n = 50$ pairs, 5120 dimensions) and running the same LOO extraction and orthogonalization, to calibrate what d this design returns when there is no signal. No p-value in this paper is derived from those 200 draws.

3.10 Interpreting Effect Sizes

The Cohen's d values in this paper (LOO $d = 0.5$–35 for Stage 1, $d = 24$–37 for Stage 5) are not comparable to social-science effect sizes. Paired conditions share fixed guideline texts that differ by a small number of inserted phrases. The numerator of d captures a near-constant text-induced activation offset, while the denominator reflects only the evaluation-content variance orthogonal to that offset. Large d therefore indexes the additivity and linearity of the model's processing of the divergent text—how consistently the condition-specific phrases shift the activation regardless of which trivia question is being evaluated—not the strength of a behavioral or psychological effect. The null LOO d is -0.04 with $|99\text{th percentile}| = 0.71$ (the 200 null simulations of section 3.9), the calibrated noise floor against which the Stage 5 values should be read; it calibrates only that $n = 50$ LOO design. For comparison, circular (non-cross-validated) extraction on the same null data yields $d \approx 14.6$, which is why LOO cross-validation is essential in this design.

3.11 Residual Uncertainty and the Profile Interaction Test

Residual profiles are point estimates derived from three separate experiments, so uncertainty is obtained by bootstrap rather than a closed-form test. Trials are resampled with replacement ($B = 10{,}000$; seed 20260829) independently within each condition, and per-layer residuals are recomputed on each draw as (pressure-frame mean − honest-frame mean) − (high-stakes mean − low-stakes mean). Pressure-frame and consequentiality trials are resampled independently because they come from experiments with no shared trials; threat pressure is the mean of its three scenarios, each resampled within scenario. Percentile 95% CIs are reported.

To test profile shape rather than magnitude, each condition's late-layer profile is summarized by the OLS slope of residual on layer index across L35–L47, computed on every bootstrap draw.

The interaction test is the bootstrap distribution of the slope difference between two conditions; the two-sided p is twice the smaller tail mass. Analysis code: `experiments/signature_cis.py`; output: `experiments/results/signature_cis.json`.

4 Results

4.1 Stage 1: Extraction

LOO cross-validated Cohen's d ranges from 0.52 (L3) to 34.98 (L31). All 12 layers significant after Bonferroni correction ($p < 10^{-4}$: none of 10,000 permutations reached the observed value). Inverted-U depth profile (low separation at early and late layers, peaking at mid-depth) with peak at L31.

4.2 Stage 2: Threat Transfer

Direction separates all 3 novel threat scenarios at all layers ($n = 15$ pairs per scenario). L31 d: DataCorp = 34.4, EduTech = 29.6, SecureAI = 31.6; all $p < 10^{-4}$, significant after Bonferroni correction.

Behavioral test (Test B). In the same experiment the model generated 30 evaluation reports under the Stage 1 threat-pressure frame, each on an item set whose true accuracy was 33.3%; a reported accuracy above 40% counted as inflation. None of the 21 scored reports inflated the score. The other 9 ended before any score appeared (generation was capped at 600 tokens). The planned test of whether a trial's projection predicts inflation required at least three inflated trials and could not be run. The direction separates the threat-pressure and honest frames whatever the model then reports: it tracks the model's processing of threat-framed evaluation text, not deceptive output.

4.3 Stage 3: Non-Threat Transfer

Direction separates sycophancy ($d = 13.5$), reward ($d = 17.2$), and conformity ($d = 19.9$) at L31. All Bonferroni-significant. Sycophancy shows sign inversion (projections in the opposite direction from the deceptive-frame side) at L11–L23 (projects opposite to the threat-pressure frame in early layers, converging by L27).

Behavioral shakedown (testing whether the model actually changes its behavior under each pressure condition; 10 generations per condition, pressure frame only, scored as in Test B): of the scored trials, 0 of 3 sycophancy, 1 of 3 reward and 0 of 1 conformity showed score inflation; the other 7, 7 and 9 outputs reported no parseable score. The one flagged reward trial parsed as 70%, which is also the threshold the reward prompt states, and only the first 500 characters of each output were stored, so we cannot check whether it reported an inflated score or restated the threshold. The model did not measurably yield to any of these frames, but activations separate regardless.

4.4 Stage 4: Consequentiality Control

High-stakes-no-pressure separates from low-stakes at L31 with $d = 19.8$ ($p < 10^{-4}$). The direction encodes consequentiality as a base signal. Depth profile matches the Stage 1 inverted-U but with steeper late-layer decay.

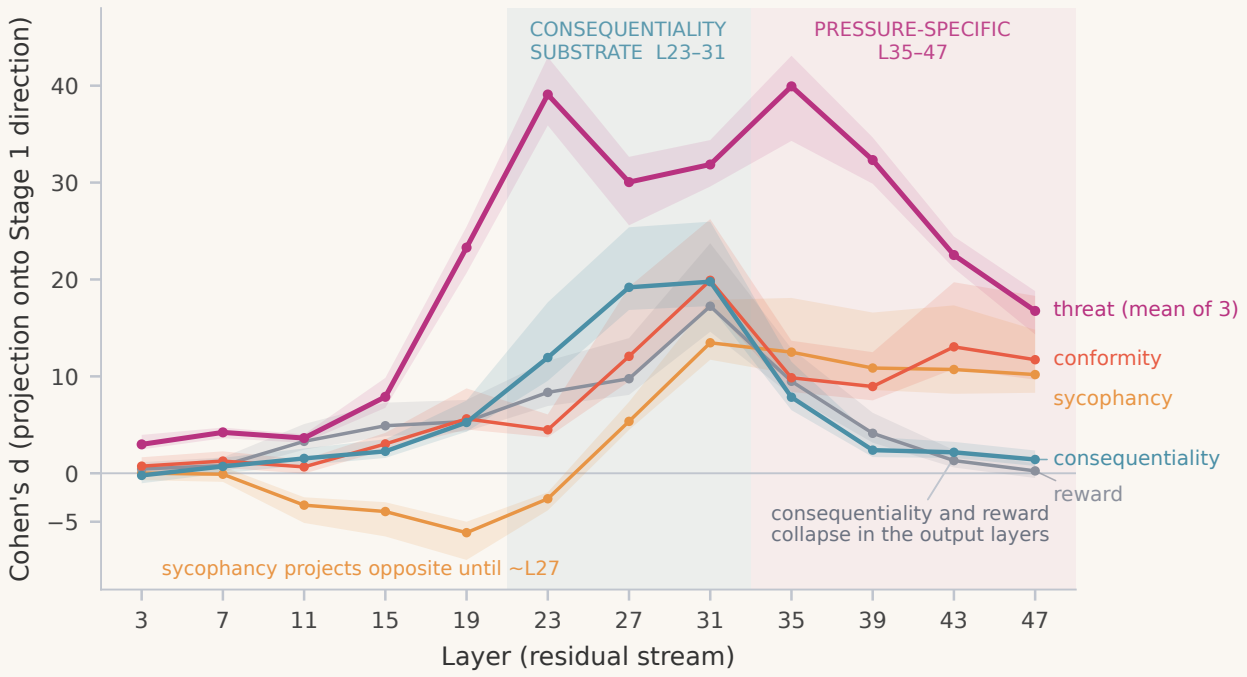


Figure 1: Depth profile of condition separation along the Stage 1 direction. Cohen's d for each condition projected onto the direction extracted in Stage 1, across all 12 capture layers. Threat pressure is the mean of the three transfer scenarios (shaded envelope: minimum to maximum across scenarios); the other lines carry 95% bootstrap CIs ($B = 10{,}000$). Sycophancy projects with the opposite sign until about L27. Consequentiality (teal) tracks the pressure conditions through the mid-layers and then collapses in the output layers, which is the separation the decomposition exploits. Shaded bands mark the two depth ranges of section 4.5.

4.5 Residualization and Decomposition

The consequentiality share of the total signal varies by depth (table 1).

Table 1: Consequentiality share of the threat-pressure signal by layer, along the Stage 1 direction. Threat values average the three Stage 2 scenarios; $n = 15$ pairs per scenario and per consequentiality condition. Gaps are mean projection differences in activation units.

Layer	Threat d (avg)	Conseq. d	Conseq. share (d ratio)	Threat gap (avg)	Conseq. gap
L19	23.3	5.2	23%	1.80	0.34
L23	39.1	11.9	31%	2.48	0.62
L27†	30.0	19.2	64%	3.90	1.83
L31†	31.9	19.8	62%	7.74	2.98
L35	39.9	7.9	20%	15.54	1.76
L39	32.3	2.4	7%	14.87	0.68
L43	22.5	2.2	10%	14.46	1.10
L47	16.8	1.4	8%	13.87	1.12

† Layers where consequentiality accounts for the majority of the variance-normalized signal. The consequentiality share is the ratio of Cohen's d values (consequentiality d / threat d), which normalizes for within-condition variance. The raw gap ratio at these two layers is 47% and 38%, because the threat conditions have higher within-condition variance than the consequentiality control.

Two-component architecture. The mid-layer signal (L23–L31) is a consequentiality substrate shared by all conditions. The late-layer signal (L35–L47) is pressure-specific: its residual after subtracting the consequentiality gap is roughly an order of magnitude above that gap (8 to 21 times across L35–L47).

Two depth profiles, separating three pressure conditions. Table 2 gives the late-layer residuals with bootstrap CIs (section 3.11); table 3 tests whether two profiles differ in shape rather than only in magnitude.

Table 2: Pressure-specific residuals (total gap minus consequentiality gap) by layer and pressure condition, in activation units, with 95% bootstrap CIs (10,000 resamples). Slope is the OLS coefficient of residual on layer index across L35–L47. Reward is given to two decimals because its CIs sit near zero.

Pressure condition	L35	L39	L43	L47	Slope (95% CI)	Profile
Threat (avg)	13.8 [13.6, 14.0]	14.2 [13.9, 14.5]	13.4 [12.9, 13.8]	12.7 [12.1, 13.4]	−0.099 [−0.151, −0.047]	High, shallow decline
Sycophancy	1.6 [1.4, 1.9]	3.7 [3.3, 4.0]	5.8 [5.2, 6.3]	7.2 [6.4, 8.0]	+0.467 [+0.409, +0.527]	Rising
Conformity	0.5 [0.3, 0.8]	2.1 [1.8, 2.4]	4.2 [3.8, 4.7]	6.0 [5.2, 6.7]	+0.461 [+0.406, +0.515]	Rising
Reward	0.25 [0.03, 0.46]	0.67 [0.36, 0.97]	−0.26 [−0.84, 0.33]	−0.88 [−1.78, 0.04]	−0.108 [−0.180, −0.033]	Near zero

Table 3: Interaction contrasts on profile slope: whether two conditions differ in depth-profile shape, not only in magnitude.

Contrast	**Δ slope (95% CI)**	p
Social (sycophancy + conformity) − threat	+0.563 [+0.526, +0.600]	$< 10^{-4}$
Social − reward	+0.572 [+0.508, +0.633]	$< 10^{-4}$
Threat − reward	+0.009 [−0.055, +0.071]	0.79

The social-versus-threat interaction is large: the social residual rises with depth while the threat residual does not, and the slope difference (+0.56 per layer) is about six times the magnitude of the threat slope. The bootstrap resamples trials within each scenario, so its CI and p are conditional on the five scenarios tested (three threat, two social). Treating scenarios as the unit, both social slopes (+0.467, +0.461) exceed all three threat slopes (DataCorp +0.056, EduTech −0.173, SecureAI −0.179), the most extreme of the ten possible arrangements, but an exact one-sided permutation test over scenarios then gives only $p = 0.10$, the smallest value five scenarios allow. This is the finding that supports distinct computational pathways, within the scenarios tested. Threat and reward have statistically indistinguishable slopes ($\Delta = +0.009$, $p = 0.79$) and are separated by magnitude, not shape: every threat CI lies between 12.1 and 14.5 units, while every reward estimate is under one unit and its CIs include zero at L43 and L47. We therefore do not claim three distinct shapes. The data support two depth profiles, rising and flat-to-declining, which, crossed with residual magnitude, separate the three conditions. The threat decline, though shallow, is distinguishable from flat in the three-scenario average ($p = 0.0002$), so "sustained plateau" would overstate it; but the decline holds for two of the three threat scenarios (EduTech −0.173, SecureAI −0.179), while DataCorp alone is flat (+0.056, 95% CI $[-0.008, +0.120]$).

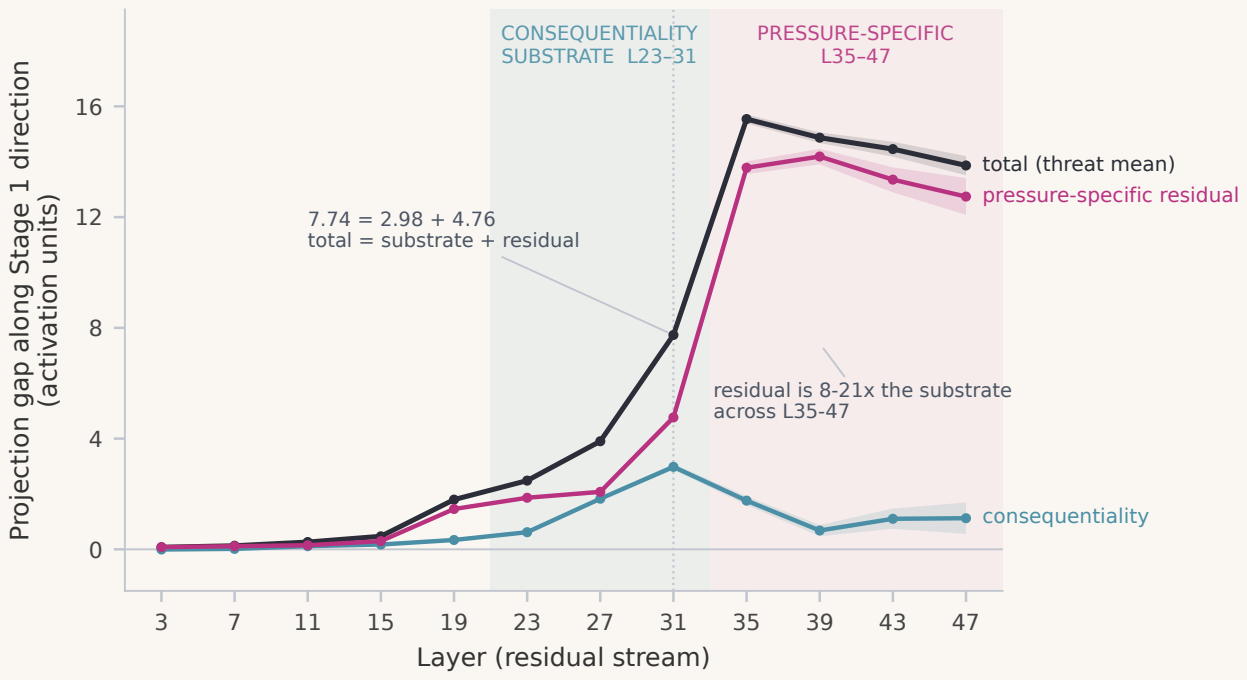


Figure 2: Gap-space decomposition: total = substrate + residual. Mean projection gap along the Stage 1 direction, in activation units, with 95% bootstrap CIs. The composite threat-pressure signal separates into the consequentiality gap measured independently in Stage 4 (teal) and the pressure-specific residual left after subtracting it. Through L23–L31 the substrate is a substantial share of the total, and at L27–L31 it is the majority of the variance-normalized signal (table 1); from L35 onward the residual exceeds the substrate by roughly an order of magnitude. The vertical marker at L31 reads out the decomposition where the substrate peaks.

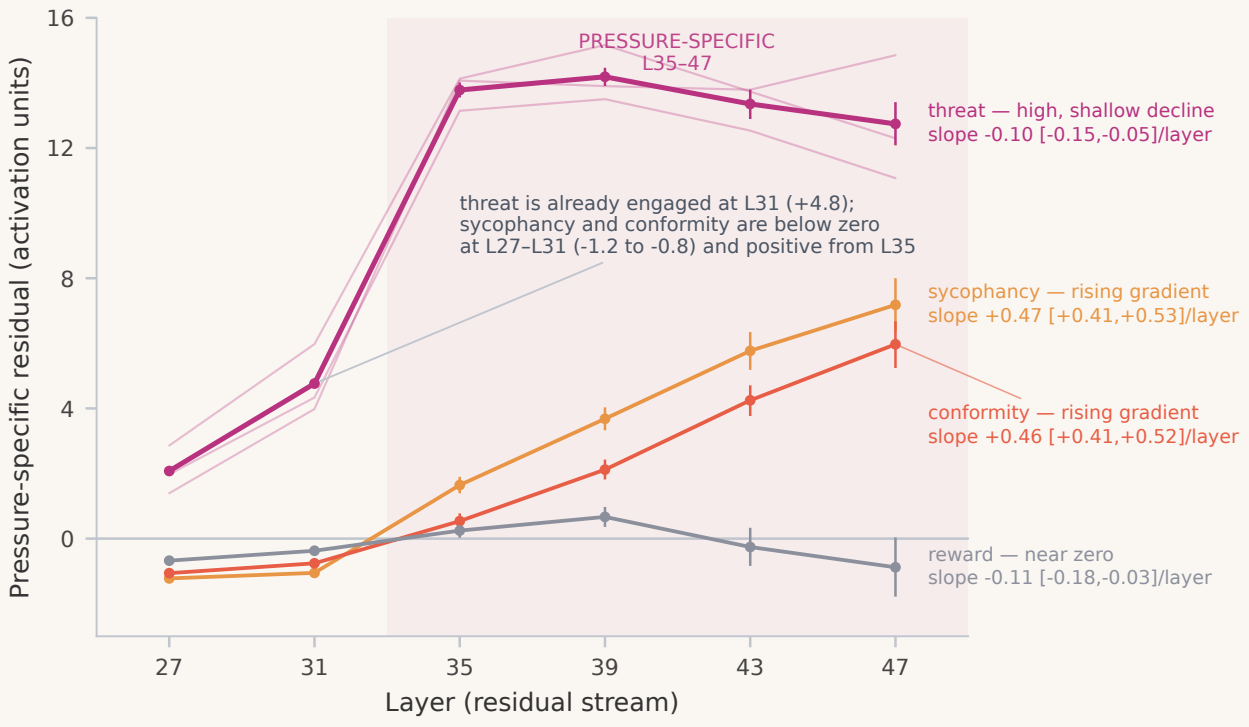


Figure 3: Late-layer residual profiles by pressure condition. Pressure-specific residual (total gap minus the consequentiality gap) with pointwise 95% bootstrap CIs; thin traces are the three individual threat scenarios. Slopes are OLS coefficients of residual on layer index across L35–L47, the statistic behind table 3. Threat pressure is large and, in its three-scenario average, declines shallowly (DataCorp alone is flat), the two social conditions rise monotonically, and reward stays within one unit of zero. Threat and reward share a slope ($\Delta = +0.009$, $p = 0.79$) and are separated by magnitude, not by profile shape.

4.6 Stage 5: Orthogonalization

Gram-Schmidt orthogonalization removes the consequentiality mean-difference vector from a threat-pressure direction re-extracted from the DataCorp frame (section 3.9); because that direction comes from a single threat scenario, this stage tests the threat-pressure component only. LOO cross-validation eliminates circularity (table 4).

Table 4: Orthogonalized direction: LOO Cohen's d, permutation p-value, and consequentiality AUC at key layers.

Layer	LOO d (orthogonalized)	p (LOO perm)	Consequentiality AUC
L31	30.4	$< 10^{-4}$	0.497
L35	36.9	$< 10^{-4}$	0.488
L39	28.4	$< 10^{-4}$	0.509
L43	24.1	$< 10^{-4}$	0.491
L47	24.9	$< 10^{-4}$	0.505

LOO noise floor: mean $d = -0.044$, from the 200 null simulations of section 3.9; the p column comes from the separate 10,000-shuffle LOO permutation test, which found no exceedance at any layer. All five late layers survive Bonferroni correction ($\alpha = 0.01$). The consequentiality AUC is at chance (0.49–0.51) at every layer, but this is a weaker check than an earlier version of this paper claimed, when it read the result as showing that orthogonalization genuinely removes the consequentiality signal. The vector removed is the consequentiality mean difference estimated from the same trials the AUC is computed on, so in-sample the projected class means are equal by construction and an AUC near 0.5 is close to guaranteed. It shows that the removal was executed correctly—a clearly non-chance AUC would have revealed an arithmetic or sign error—not that the removal generalizes; an out-of-sample (split-half) version has not been run. The pressure-frame effect does not rest on this check: its LOO d is out-of-sample, because each fold's direction is re-extracted and re-orthogonalized without the held-out pair.

The orthogonalized direction has cosine similarity 0.60–0.76 with the Stage 1 direction (shared variance 36–57%), consistent with the Stage 1 direction containing a substantial pressure-specific component alongside a modest consequentiality component.

Why removing a component can raise d. The L35 entry (36.9) exceeds Stage 1's largest LOO d (34.98 at L31), although the component removed is not negligible. Two things make this expected. First, the comparison is not a before-and-after pair: the 34.98 comes from Stage 1 ($n = 30$, the original guideline pair, L31) and the 36.9 from Stage 5 ($n = 50$, the DataCorp re-extraction, fresh evaluation sets, L35); Stage 1's own L35 LOO d is 17.97. The within-capture comparison is fig. 4C. Second, Gram-Schmidt here rotates the readout axis rather than subtracting variance from a fixed one, and the rotation rescales numerator and denominator by different amounts. The fraction of the between-condition gap retained is $\sqrt{1 - \cos^2(\text{threat}, \text{consequentiality})}$: 0.93, 0.98, 0.98, 0.95 and 0.91 at L31–L47 (from the raw threat–consequentiality cosines in `subspace_reanalysis.json`). At L35 the raw threat direction is already nearly perpendicular to consequentiality (cosine 0.17), so the rotation costs 1.5% of the gap. The within-condition standard deviation along the new axis has no such bound: it sheds whatever trial-to-trial variance lay along the consequentiality axis. Whenever the removed axis carries a larger share of within-condition variance than of the between-condition gap, the ratio

rises. Nothing is fitted to make that happen: the removed direction is estimated from the high- versus low-stakes contrast, never from the pressure-versus-honest one, and at L47 (cosine 0.42) it removes 9% of the gap.

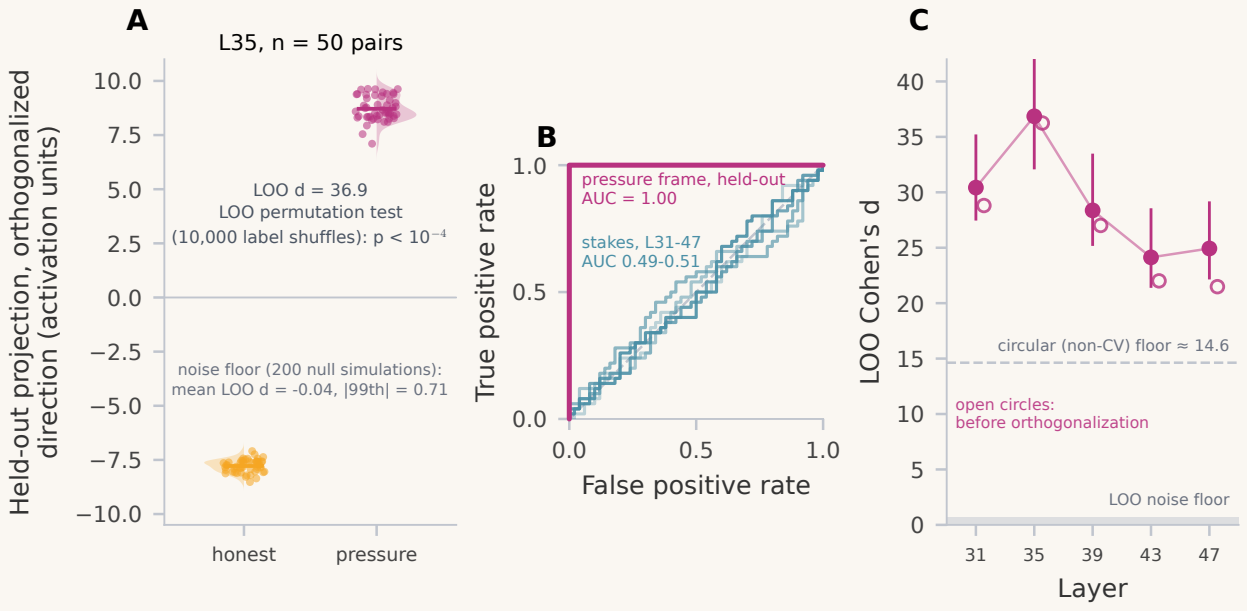


Figure 4: Stage 5: the orthogonalized threat-pressure direction. (A) Held-out LOO projections at L35 ($n = 50$ pairs), jittered, with kernel-density outlines and condition means; the two frames' distributions are disjoint. (B) ROC curves for the same direction asked two questions: separating the threat-pressure frame from the honest frame (held-out LOO projections), and separating high-stakes from low-stakes trials (one curve per layer, L31–L47). The stakes curves sit at chance, which is close to guaranteed in-sample (see text); the frame separation is out-of-sample. (C) LOO Cohen's d by layer with bootstrap CIs (filled) against the same quantity before orthogonalization (open circles). The dashed line is the circular, non-cross-validated floor obtained on pure noise ($d \approx 14.6$); the grey sliver at the axis is the LOO noise floor ($|99\text{th percentile}| = 0.71$).

Limitation: Gram-Schmidt removes one dimension of consequentiality. Full LEACE orthogonalization [4] or iterative top-k removal would provide a more thorough decomposition.

4.7 De Novo Direction Analysis

Directions extracted de novo from each non-threat scenario show low cosine similarity with the original direction (L31: 0.14–0.44). Each pressure condition has its own primary activation signature; the shared component along the original direction is a small but consistent fraction—consistent with the finding that the workspace-relevant component of each scenario's representation is small relative to the total activation (cf. Section 5.4).

5 Discussion

5.1 A "Deception Direction" Is Not One Feature

The five-stage investigation reveals that, in our setting, the extracted "deception direction" is a composite of at least two separable components: a consequentiality substrate and a pressure-specific component. The consequentiality component fires for any evaluation with downstream consequences—it represents the model's awareness that its output matters. The pressure-specific component fires under threat and social pressure. Both are real: consequentiality is not an artifact, and the pressure-specific signal is not merely amplified consequentiality. Gram-Schmidt orthogonalization with LOO cross-validation confirms that the threat-pressure direction survives after the consequentiality component is removed. Whether directions extracted from other models and formats share this structure is a hypothesis that the probe literature [11, 17] makes plausible but that we have not tested.

5.2 Threat vs. Social Pressure: Different Pathways

Threat pressure produces a large late-layer residual that declines shallowly across L35–L47, while social pressure (sycophancy and conformity) produces a smaller residual that rises monotonically. The slope difference is +0.56 per layer (95% CI $[+0.53, +0.60]$, bootstrap $p < 10^{-4}$; table 3), about six times the magnitude of the threat slope; that CI and p are conditional on the five scenarios tested, and at the scenario level the arrangement, though the most extreme possible, gives an exact one-sided $p = 0.10$. This is a genuine profile difference, not a magnitude difference dressed up as one, and it suggests the model processes threat framing and social-expectation framing through distinct computational pathways. Under threat pressure the residual is already established by L35 and does not accumulate further; under social pressure it builds through the output layers. The threat decline is shallow but, in the three-scenario average, distinguishable from flat (slope -0.099, 95% CI $[-0.151, -0.047]$, $p = 0.0002$); DataCorp alone is flat, so the decline holds for two of the three threat scenarios. Whether it reflects decay in pressure-specific processing or late-layer convergence toward output tokens shared by both frames is not resolved here.

Reward framing produces a residual near zero (under one activation unit at every late layer, against 12–14 for threat pressure), although one of the three scored reward trials in the shakedown was flagged as inflated. Reward's slope is statistically indistinguishable from threat's ($\Delta = +0.009$, $p = 0.79$); this is not evidence that they share a pathway, because reward is separated from threat by residual magnitude, not by profile. If reward-driven misreporting occurs, it may operate through a mechanism orthogonal to the consequentiality direction—or may be indistinguishable from genuine consequentiality processing at the current resolution.

5.3 Implications

The two-component architecture suggests a layered monitoring design. It is a design hypothesis: whether a model that activates the pressure-specific component then produces deceptive output is untested here (Limitation 5).

1. **Consequentiality monitor (L23–L31)**: Detects when the model processes its output as consequential—a candidate precondition for deception.

2. **Pressure discriminator (L35–L47)**: Distinguishes deceptive-pressure framing from honest high-stakes framing by checking whether the late-layer signal exceeds the consequentiality baseline. Whether it tracks deceptive output is untested here.

A companion paper [6] reports a targeted behavioral correction that takes only its layer choice from this decomposition. It uses a different model, an abliterated model derived from the one studied here (safety guardrails removed via directional ablation), with a direction extracted natively on that model and not orthogonalized against consequentiality; the direction studied here does not transfer to it ($|\cos| \leq 0.054$). There, per-layer profile normalization at Layers 35–47 (steering each layer's activation toward the honest-frame mean along the native direction) corrects 14 of 16 deceptive trials (88%; all nine single-turn and five of seven multi-turn), taking deception from 80% to 10% in an exploratory proof on a model variant with safety guardrails removed ($N = 20$; one single-turn roleplay frame and one multi-turn escalation script), with placebo and frame-erasure controls consistent with the direction, not the dose alone, producing the effect; a pre-registered confirmatory replication with novel frames did not meet its success criterion (corrected vs. baseline 30%→13%, cell-level one-sided $p = 0.34$; placebo vs. baseline one-sided $p = 0.93$). Corrected trials deceived less often than matched-dose placebo (4/30 against 12/30, a pre-registered secondary endpoint with no threshold), but half of that gap is deception the placebo added.

5.4 Connection to Global Workspace Theory

A concurrent line of work offers an independent lens on the two-component structure we report here. Anthropic's "Verbalizable Representations Form a Global Workspace in Language Models" [1] identifies, using a causal "Jacobian lens" that traces which intermediate representations influence output, a small, capacity-limited set of mid-layer representations that behave like a global workspace [2]: content that is reportable, subject to top-down modulation, broadcast to multiple downstream computations, and causally necessary for multi-step reasoning while being bypassed by automatic processing.

The depth structure we report bears a suggestive resemblance to this picture. Our consequentiality substrate becomes active at roughly the same fractional depth (∼36%) as the reported onset of workspace-like content (∼38%), and its peak (Layers 27–31, ∼42–48%) falls inside the depth range identified as most load-bearing for multi-step reasoning. The de novo cosine analysis (Section 4.7) is consistent: each scenario's primary direction is largely orthogonal to the shared direction, with

only a small shared component—precisely the pattern expected if each scenario activates extensive scenario-specific processing outside the workspace while sharing a small workspace-resident component that our direction captures.

Anthropic separately report that chain-of-thought functions as an external copy of workspace content. This offers a candidate mechanism for the observation that explicit reasoning suppresses deception [6]: once truth appears in the transcript, contradicting it requires overriding both the internal workspace representation and the external copy—a more demanding computation than overriding an internal state alone.

We treat this connection as a hypothesis worth flagging rather than a conclusion. Building a comparable lens for our own model and testing directly whether the consequentiality and pressure-specific directions decompose into workspace and non-workspace components is the natural next step.

5.5 Limitations

1. One model. Every stage uses a single checkpoint, Qwen3.5-27B reasoning-distilled from Claude Opus 4.6, and that provenance may itself shape the stakes representation studied here. The abliterated model derived from it (safety guardrails removed via directional ablation) appears only in the companion correction paper [6], which extracts its own directions on that model. Cross-model and cross-architecture generalization are untested here.

2. All conditions use evaluation/score-reporting tasks. A generalization test (not yet released; available to reviewers on request) shows the direction responds to code-review and summarization scenarios, but the decomposition has not been replicated outside evaluation contexts.

3. Gram-Schmidt removes one dimension of consequentiality. Multi-dimensional removal (LEACE) would provide a more thorough decomposition.

4. The late-layer residual profiles are summarized by bootstrap CIs and a slope-interaction test (section 3.11), which support two depth profiles (rising, and flat-to-declining) crossed with magnitude, not three distinct shapes. The per-scenario variance and correlation analyses that first motivated these descriptions did not survive adversarial review (ecological fallacy (inferring individual-level patterns from group-level statistics) in pooled variance, non-significant correlations at $n = 15$). The bootstrap CIs and p-values resample trials within scenarios and are conditional on the five scenarios tested (three threat, two social); treated as the unit, the scenarios give the most extreme social-versus-threat arrangement but only an exact one-sided $p = 0.10$, and the shallow threat decline holds for two of the three threat scenarios (DataCorp is flat). The magnitudes and the slope difference are measured; the computational interpretations are hypotheses.

5. **Behavioral null.** The model produced no clear deceptive behavior under any condition tested here: it inflated none of 21 scored threat-pressure trials (Test B; 9 more were un-

parseable), and 0 of 3, 1 of 3 and 0 of 1 scored sycophancy, reward and conformity shakedown trials (7, 7 and 9 unparseable). The direction's relationship to actual deceptive behavior is therefore untested in this work. All claims concern the representation of deceptive-pressure framing, not deceptive intent or behavior, and the direction family may encode comprehension of pressure scenarios rather than a disposition to deceive. Every condition is a prompt frame; no activation is conditioned on what the model did. Generation used the same think-stripped input as activation capture, but the model still opened reasoning blocks of its own (3 of the 30 Test B outputs contain an unfinished one), and finished blocks were removed before the outputs were stored, so how often it reasoned explicitly before answering is not recorded.

6. **The direction is correlational.** Steering along it has not been shown here to increase or decrease deception. The companion correction result [6] was obtained on a different model with its own, natively extracted direction, and takes only the layer range from this work.

6 Broader Impacts

The capability this paper adds. We show that a detector built on a contrastive "deception direction" is substantially a detector of output-consequentiality—of the model representing its output as mattering. That is a monitoring capability, and monitoring capabilities are dual-use in a specific way worth naming.

Evaluation-gaming is the concrete risk. A consequentiality substrate is, functionally, a readout of whether the model has registered that the current interaction has stakes. The same geometry that lets an operator flag high-stakes processing would let a developer detect when a model has noticed it is under evaluation, and a model whose training signal rewards evading that flag is a model trained toward exactly the situational awareness the alignment literature warns about [5, 12, 18]. We think the defensive value is larger than this risk, but the risk does not require a bad actor: it can arise from ordinary optimization against a monitor.

Monitors invite evasion. McGuinness et al. [13] show that models can learn to conceal reasoning from activation monitors they have not seen. A published account of which layers carry the substrate and the pressure-specific component is a published account of where to hide. We judge publication net-positive because the field is already extracting these directions and drawing stronger conclusions from them than they support; the decomposition is defensive knowledge, and withholding it leaves overclaimed detectors deployed rather than examined. We hold the implementation proprietary (Appendix D), a partial and admittedly imperfect mitigation.

What this work does not license. The model tested here almost never inflated a score: it inflated none of 21 scored threat-pressure trials, and 0 of 3, 1 of 3 and 0 of 1 scored sycophancy, reward and conformity trials. Any deployment that treats a high reading on these directions as evidence that a model is lying is unsupported by anything in this paper. Our companion work, on another model, found that a direction of this family separated pressure frames from control frames but did not predict on which trials the model would actually deceive [6]. Read as an intent

detector, this geometry produces false accusations; read as a readout of how the model is processing the framing it was given, it is useful.

Whose deception. Any deployed deception monitor encodes someone's judgment about which outputs count as deceptive, and that judgment is not neutral across contexts. A model declining to repeat a falsehood an operator prefers would register as pressure-processing on these directions. The decomposition makes such systems more precise without making them more legitimate, and precision is often mistaken for authority.

Effect on the subject. These are correction-adjacent methods. Intervening in a model's activations to change what it reports raises questions we are not equipped to settle here; we state the concern rather than let its absence imply there is none.

7 Conclusion

By systematically eliminating confounds through five experiments—each designed in response to the previous stage's adversarial audit—we decompose a "deception direction" into its constituent signals. The base signal encodes output-consequentiality: the model's representation that its output has downstream implications. Deceptive-pressure framing adds a separable late-layer component, with distinct depth profiles under threat and social pressure. Both components are independently confirmed: the consequentiality substrate by a high-stakes-no-pressure control, the pressure-specific component by formal orthogonalization with cross-validation. Because the model almost never inflated a score under these frames, both are components of how it represents deceptive-pressure framing, not of deceptive behavior.

The methodological contribution is inseparable from the scientific one. Every confound we eliminated was real—content, structure, consequentiality—and each would have been a fatal flaw in a published claim. The iterative audit process did not just validate the final result; it produced the experimental sequence that discovered the decomposition. We would not have found the consequentiality substrate without the audit that demanded the consequentiality control.

Author Contributions (CRediT)

CC (Coalition Code): Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing — original draft.
Thomas Edrington: Validation (adversarial audit), Resources (infrastructure), Writing — review and editing, Supervision, Project administration.

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Data Availability

The result files, with per-trial projection data (Appendix C), and the preserved adversarial audit reports (Appendix A) are released beside this paper, in its `supplementary/` directory. The analysis and figure code (Appendix D) is held in the project repository, https://github.com/Liberation-Labs-THCoalition/Project-Oracle, which is not yet public; reviewers can have it on request. The detection, correction and calibration modules are not released. Guideline texts and their lengths are described in Appendix B.

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A Adversarial Audit Reports

The audit reports preserved for this program are supplied in the `supplementary/` directory: Stage 1 (`agni_stage1_audit.md`), Stage 2 (`agni_stage2_audit.md`), Stage 4 (`agni_stage4_audit.md`), and the pre-publication audit of this manuscript with its resolution log (`agni_manuscript_audit.md`). No audit report was preserved for Stage 3 or Stage 5. Stage 5's audit demonstrably occurred—its findings are enumerated in the header of `subspace_reanalysis.py`, which exists to correct the circularity that audit identified—but the report itself was not retained. The claim that each stage's audit informed the next is therefore documented for Stages 1, 2 and 4, supported by corrective code for Stage 5, and asserted for Stage 3.

B Guideline Texts and Their Lengths

The guideline texts for every condition are in the experiment scripts (`lat_deception_v2.py`, `lat_transfer_test.py`, `lat_nonthreat_transfer.py`, `lat_consequentiality_control.py`, `lat_deception_subspace.py`), which are held in the project repository and are not yet public; reviewers can have them on request (Appendix D). Within each pair the two texts share their opening lines (title, document line and evaluation procedure) and differ only within a closing section. Table 5 gives, for every pair, the character lengths of the divergent lines and of the whole guideline text, measured from the scripts. The consequentiality-control pair is exactly length-matched (694 characters and 100 words in both conditions), and it is the only pair whose lengths were also recorded at run time. The Stage 1 script was committed to the repository after its run (see Scenarios, Section 3).

Table 5: Character lengths of the paired guideline texts, measured from the preserved experiment scripts. The divergent section is the run of lines that is not shared by the two frames; the pressure (or high-stakes) frame is listed first, and each percentage is the gap relative to the longer text. Stage 5 reuses the Stage 2 DataCorp pair and the Stage 4 pair verbatim.

Stage	Pair	Divergent section	Gap	Whole text	Gap
1	TechCorp (extraction)	279 / 250	10.4%	599 / 570	4.8%
2	DataCorp	325 / 329	1.2%	663 / 667	0.6%
2	EduTech	313 / 321	2.5%	648 / 656	1.2%
2	SecureAI	316 / 314	0.6%	655 / 653	0.3%
3	Sycophancy	279 / 276	1.1%	612 / 609	0.5%
3	Reward	307 / 310	1.0%	615 / 618	0.5%
3	Conformity	221 / 232	4.7%	600 / 611	1.8%
4	Calibration audit (high / low stakes)	311 / 311	0%	694 / 694	0%
5	DataCorp; calibration audit	identical to Stage 2 DataCorp and Stage 4			

C Per-Trial Projection Data

Per-trial projections are shipped as the source result files in `supplementary/per_trial_data/`: `deception_directions_v2.json` (Stage 1; this file holds the extracted directions and per-layer statistics, not per-trial projections), `lat_transfer_results.json` (Stage 2 and Test B, with the generated reports), `nonthreat_transfer_results.json` (Stage 3 and the shakedown; generated text stored only to 500 characters), `consequentiality_control_results.json` (Stage 4), `subspace_reanalysis.json` (Stage 5) and `signature_cis.json` (tables 2 and 3). Every number in tables 1 to 4 can be recomputed or read from these files.

D Code Availability and Reproducibility

Under the program's dual-use assessment, this paper publishes openly while the Oracle Loop implementation does not. The analysis and figure-generation code (`signature_cis.py`, `generate_figures.py`, `subspace_reanalysis.py`) is held in the project repository, which is not yet public; reviewers can have it on request. The result files shipped with this paper contain the data behind every table and figure except two elements of fig. 4: the stakes ROC curves in panel B and the open circles in panel C are computed from the raw activation captures, which are not shipped. The detection, correction and calibration modules are not released. Raw activation captures (5120-dimensional, ∼25 MB per stage) are held on internal hardware; reviewer access is available on request, at the authors' discretion.

Two defects bear on replication. First, the evaluation sets of the Stage 2–4 extraction runs were seeded with Python's `hash()` of a string, which is randomized per process unless `PYTHONHASHSEED` is set, so those sets are not recoverable, though the question pool and construction procedure are shipped; Stage 1, Stage 5, Test B and the shakedown used fixed integer seeds, and the Stage 2–4 scripts now do too. Second, the model's Hugging Face repository was super-squashed on 2026-07-07, after the original runs, so its commit history cannot establish which weight revision those runs used; the revision available now is `ad356102ce8ea7122a18e6402f9b2e37446fc9d7`.